Customer Segmentation & Campaign ROI Dashboard

Power BI Portfolio Project

Prepared by: Alek

Data Analytics Portfolio | Power BI, DAX, Power Query

Executive Summary

This project analyzes 2,240 customers using a real-world marketing dataset to answer a core business question: which customers are most valuable, and which marketing campaigns are worth repeating? The result is a five-page interactive Power BI dashboard covering customer segmentation, product performance, campaign effectiveness, and purchasing channel behavior.

The dashboard was built end-to-end: raw data cleaning and reshaping in Power Query, a working data model with proper table relationships, over 20 custom DAX measures, and five fully designed, styled, and interactive report pages.

Business Objective

The business ran six marketing campaigns and wanted to understand two things: which customer segments generate the most value, and which campaigns actually converted. This dashboard was designed to answer that directly, structured around five guiding questions:

- Who are our customers, and what does the overall business look like? (Overview)
- Which customer segments — by income, marital status, and family structure — drive the most revenue? (Customer Segmentation)
- Which product categories perform best, and for whom? (Product Performance)
- Which campaigns were effective, and who responded to them? (Campaign Effectiveness)
- How do customers prefer to purchase — web, store, or catalogue? (Channel Behavior)

Data Source & Preparation

The dataset is a customer-level marketing table (2,240 rows) with demographic fields, campaign response flags, product spend, and purchase channel counts. Several fields required cleaning before analysis:

- Marital status and education were stored as one-hot encoded columns (e.g., marital_Married, marital_Single) and were consolidated into single categorical columns using conditional logic.
- The six campaign flags (AcceptedCmp1–5, Response) were unpivoted in Power Query into a dedicated Campaigns table with one row per customer per campaign, enabling proper campaign-level analysis.
- Income Band groupings were created to support segmentation analysis without exposing raw income values.
- A relationship was established between the Customer table and the unpivoted Campaigns table on Customer ID.

Dashboard Structure

Page 1 — Overview

A snapshot of the overall business: total customers, total revenue, average spend, campaign acceptance rate, complaint rate, and average recency, paired with a revenue-by-category donut chart, a customer count by income band chart, and a customer tenure distribution.

Metric	Value
Total Customers	2,240
Total Revenue	~1,000,000+
Avg Spend per Customer	562.76
Overall Campaign Acceptance Rate	7.51%
Complaint Rate	0.91%
Avg Recency (days)	49.01

Page 2 — Customer Segmentation Analysis

Breaks down revenue and average spend by marital status. Married customers generate the highest total revenue of any segment, though average spend per customer is more evenly distributed — Widowed customers, despite being the smallest group, show the highest per-person average spend, highlighting the difference between segment size and segment value.

Page 3 — Product Performance

Wine is the standout category, contributing just over 50% of total category revenue on its own — more than the next several categories combined. This is a clear signal for where merchandising and promotional focus could have the greatest impact.

Page 4 — Campaign Effectiveness

Shows acceptance rate by individual campaign against the 7.51% overall baseline, campaign performance broken down by education level via a conditional-formatted matrix, and a repeat-responder rate identifying customers who accepted more than one campaign — a useful signal for future targeting.

Page 5 — Channel Behavior

Compares purchase volume across Web, Store, and Catalogue channels, examines web visit-to-purchase conversion, and breaks down deal usage and channel preference by income band and age group.

Tools & Skills Demonstrated

- Power Query: conditional columns, unpivoting, data type correction, table relationships
- DAX: CALCULATE, FILTER, SWITCH, RANKX, TOPN, DIVIDE, iterator functions (AVERAGEX, SUMX)
- Data modeling: star-schema-style relationships between a customer dimension and an unpivoted campaign fact table
- Report design: custom theme, consistent color system, conditional formatting heatmaps, synced slicers, and custom page navigation

Challenges & Key Learnings

Several real debugging challenges came up during this build, each resolved through systematic troubleshooting rather than guesswork:

- Distinguishing measures from calculated columns — an early attempt to build Marital Status as a measure caused every chart referencing it to return blank, since measures can't be used as chart-axis categories the way columns can.
- Fixing incomplete Power Query unpivots that initially dropped five of six campaign columns, leaving only one campaign represented in the data.
- Handling a measure that correctly returns blank in a filtered context (Repeat Responder Rate under a single-campaign filter) — resolved with an ISBLANK wrapper rather than treating it as a bug.
- Building consistent, on-theme formatting (custom navy/teal/gold theme, sidebar navigation with active-page highlighting) across all five report pages.

Conclusion

This dashboard demonstrates a complete analytics workflow — from messy, one-hot-encoded raw data through to a polished, interactive, five-page business intelligence report. It answers concrete business questions about customer value, product performance, campaign ROI, and channel behavior, and reflects hands-on problem-solving with real data modeling and DAX challenges along the way.